

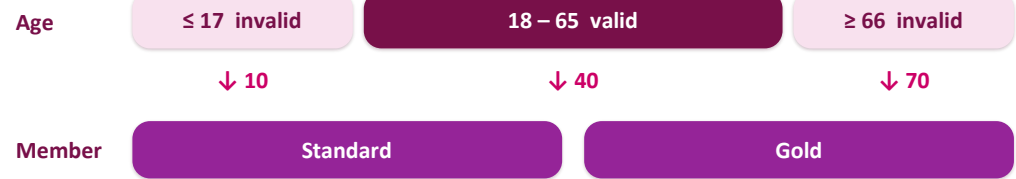
Practice Workbook

A. Worked example: decision table

Conditions / actions	R1	R2	R3	R4
Card valid?	F	T	T	T
PIN correct?	–	F	T	T
Sufficient funds?	–	–	F	T
Reject card	X			
'Incorrect PIN'		X		
'Insufficient funds'			X	
Dispense cash				X

Cash machine: card valid? PIN correct? Sufficient funds? The **full** table has $2^3 = 8$ columns. If the card is invalid, the other conditions don't matter (–); if the PIN is wrong, funds don't matter. Merging gives the **minimised** table above: **4 rules** → **4 test cases** for 100% coverage.

B. Worked example: EP and BVA

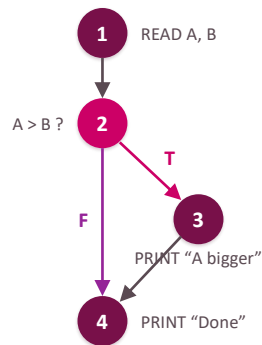


Rule: applicants aged 18–65 are eligible. Membership is Standard or Gold.

EP with Each Choice: 3 tests cover all 5 partitions, e.g., (10, Standard), (40, Gold), (70, Standard).

- **2-value BVA:** 17, 18, 65, 66 → 4 tests
- **3-value BVA:** 16, 17, 18, 19, 64, 65, 66, 67 → 8 tests

C. Worked example: statement and branch coverage

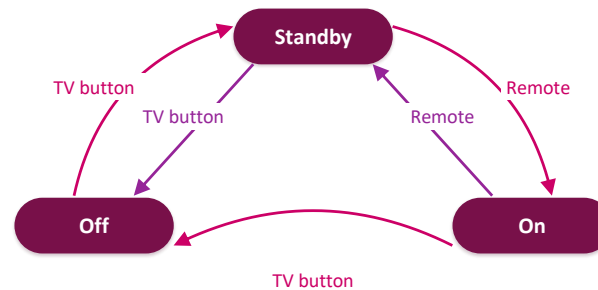


```

1 READ A, B
2 IF A > B THEN
3   PRINT "A bigger"
4 PRINT "Done"
  
```

- Test 1 (A=5, B=2): all 4 statements → **100% statement**; branches 1→2, 2→3, 3→4 but not 2→4 → **3/4 = 75% branch**
- Add test 2 (A=1, B=3) → **100% branch**

D. Worked example: state transitions



State table: Off: TV button → Standby, Remote → (invalid). Standby: TV button → Off, Remote → On. On: TV button → Off, Remote → Standby.

- **All states** (3 items): Off → Standby → On
- **Valid transitions** (5 items), one test: Off → TV Standby → Remote On → Remote Standby → TV Off → TV Standby → Remote On → TV Off
- **All transitions** (6 items): the test above + a separate test pressing Remote in Off (expect no change)

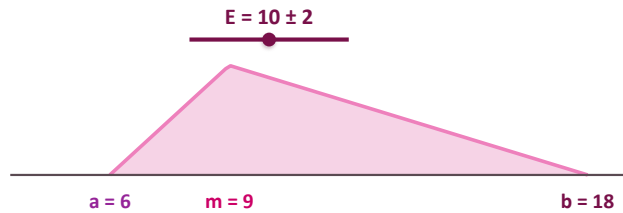
E. Exercise 1: password length

A password must be **8 to 20** characters long (inclusive).

1. List the equivalence partitions and one test value for each.
2. List the coverage items for **2-value BVA**.
3. List the coverage items for **3-value BVA**.

Answers at the back.

F. Worked example and Exercise 3: estimation



Three-point: $a = 6, m = 9, b = 18 \rightarrow E = (6 + 36 + 18) / 6 = 10$; $SD = (18 - 6) / 6 = 2 \rightarrow 10 \pm 2$ person-hours.

Exercise 3

1. Past dev:test ratio 5:3. New dev effort 250 person-days. Test effort?
2. Last three iterations took 18, 21, 24 test hours. Next iteration (extrapolation)?
3. $a = 4, m = 7, b = 16$. Three-point E and SD?

G. Exercise 2: minimum tests for 100% coverage

Rules of thumb (simple, non-nested): IF without ELSE: 1 statement / 2 branch · IF-ELSE: 2 / 2 · SWITCH with n outcomes: n / n · loop a test can enter and exit: 1 / 1.

Q1

```
READ P, Q
IF P + Q > 100 THEN PRINT "Large" ENDIF
IF P > 50 THEN PRINT "P Large" ENDIF
```

Q2

```
READ A, B, C
IF C > A THEN
  IF C > B THEN PRINT "X"
  ELSE PRINT "Y" ENDIF
ELSE PRINT "Z" ENDIF
```

How many tests for 100% statement and 100% branch coverage? Answers at the back.

H. Exercise 4: match tools to test activities

For each tool type, write the test activities it mainly supports, what it does, and an example you know.

- **Management:** planning, monitoring & control, completion · manages requirements, tests, defects · e.g., Jira + Xray
- **Static testing:**
- **Test design & implementation:**
- **Test execution & coverage:**
- **Non-functional:**
- **DevOps:**
- **Collaboration:**
- **Scalability & deployment:**

I. Worked example: ATDD

US-88 Apply a discount code

As a shopper,
I want to apply a discount code
so that I pay less.

Acceptance criteria

- valid code → 10% off
- expired code → error

Card

the medium

Conversation

how it's used

Confirmation

acceptance criteria

- **Positive:** Given a £50 basket, When I apply WELCOME10, Then the total is £45
- **Negative:** Given a £50 basket, When I apply an expired code, Then I see 'Code expired' and the total stays £50

J. Worked example: prioritisation

TC1 (High, depends on TC3) · TC2 (Medium) · TC3 (Low) · TC4 (High).

Run high-priority tests as early as possible; TC3 must run **before** TC1. One valid order: **TC4 → TC3 → TC1 → TC2** (TC3 → TC1 → TC4 → TC2 is equally valid).

K. Worked example: a defect report

- **ID / title:** DEF-214 Discount code accepted after expiry date
- **Date / author:** 23-09-2026 · J. Smith (tester)
- **Test object / environment:** Checkout service v2.3.1 · UAT, Chrome 128
- **Context:** TC-CHK-017 (acceptance criterion AC3), system testing
- **Steps:** 1. Add £50 item 2. Enter SUMMER24 (expired 31-08-2026) 3. Click Apply
- **Expected / actual:** 'Code expired', total £50.00 / code accepted, total £45.00
- **Severity / priority / status:** High · High · Open
- **References:** TC-CHK-017, US-88 · screenshot, server log

L. Answers

- **Exercise 1:** partitions ≤ 7 (invalid), 8–20 (valid), ≥ 21 (invalid), e.g., 5, 12, 25. **2-value BVA:** 7, 8, 20, 21. **3-value BVA:** 6, 7, 8, 9, 19, 20, 21, 22.
- **Exercise 2: Q1:** 1 test for statement ($P=60$, $Q=50$), 2 for branch (add $P=10$, $Q=10$). **Q2:** 3 and 3 (X, Y and Z each need their own path).
- **Exercise 3:** 1. $250 \times 3/5 = 150$ **person-days**. 2. $(18 + 21 + 24) / 3 = 21$ **hours**. 3. $E = (4 + 28 + 16) / 6 = 8$, $SD = (16 - 4) / 6 = 2 \rightarrow 8 \pm 2$.
- **Exercise 4** (suggested): static testing: reviews and static analysis. Design & implementation: test cases, data, procedures. Execution & coverage: automated runs and coverage. Non-functional: performance, security, etc. DevOps: pipeline, CI/CD. Collaboration: all activities. Scalability & deployment: test environments (VMs, containers).